CastIQ

Business Partner Tech Overview

Where everything lives | What to change | Where credentials go

Product snapshot

Product	Minnesota lake intelligence platform	
	github.com/swehrcustoms/CastIQ-App	**Reposito**
	castiq-app	**Firebase**
	Vite + React + TypeScript · Firebase · Mapbox · Anthropic Claude · Tomorrow.io	**Stack**

1. Big picture

```
User (browser / phone)
    |
Firebase Hosting  <--  built from Vite app (src/)
    |
Firebase Auth + Firestore + Storage
    |
Cloud Functions (functions/)  <--  AI, weather, DNR ingestion, guide portal
    |
External APIs: Anthropic, Tomorrow.io, Mapbox, MN DNR feeds
```

2. Frontend app (what users see)

Change UI and product screens under src/. Routes are defined in src/App.tsx.

What	Where to change it
Routes / screens	src/App.tsx
Pages (Map, Search, Lake, Login, Guide, Reports, Regs)	src/pages/
UI components	src/components/
Auth / Firebase / Firestore helpers	src/services/
Data hooks	src/hooks/
TypeScript data models	src/types/
Brand colors	tailwind.config.cjs
App entry + styles	src/main.tsx, src/index.css

Screens today

Route	Page	Purpose
/login	Login	Email/password signup & sign-in
/	MapHome	Map of lakes
/search	LakeSearch	Find lakes

Route	Page	Purpose
/lake/:id	LakeDetail	Lake overview + reports
/lake/:id/live	AILiveEngine	Live AI fishing intel
/regulations	Regulations	Fishing regulations
/submit-report	SubmitReport	Angler reports
/guide	GuidePortal	Guide business portal

Coming-soon placeholders: fisheries, contours, border waters, ice mode.

3. Backend (Cloud Functions)

Area	Folder	What it does
Function exports	functions/src/index.ts	Master list of deployed functions
Scheduled ingestion	functions/src/ingestion/	RSS, DNR lakes, water quality, ice
Data APIs	functions/src/services/	Lakes, geo search, surveys, regs, reports
AI	functions/src/ai/	Claude summarization, live intel, winterkill
Guide / contours / structures	functions/src/v12/	Guide portal, contours, similar lakes, sonar
Notifications	functions/src/background/	Alert dispatch
Claude client	functions/src/utils/anthropicClient.ts	Anthropic API wiring

Functions region: us-central1

4. Database (Firestore)

Collection names are defined in src/services/firestore.ts.

Collection	Contents
lakes	Lake records
angler_reports	User fishing reports
species_registry, lures_colors	Species & lure data
regulations, statewide_defaults, regulation_alerts, border_waters	Rules
netting_surveys, electrofishing_surveys, creel_surveys, water_quality	Surveys / science
stocking_events, winterkill_events, aeration_events	Events
users, guide_portal_users	Accounts & guides
weather_cache, live_intelligence_cache, similar_lakes	Cached AI/weather
ice_reports, ice_roads	Ice fishing
lake_contour_metadata, lake_structures, contour_update_log	Contours / structures

- Security rules: firestore.rules
- Indexes: firestore.indexes.json
- File uploads: Firebase Storage rules in storage.rules

5. Scripts & seed data

Script	Command	Purpose
Seed lakes/species/regs/reports	npm run seed:all	Load starter data

Script	Command	Purpose
Create guide account	npm run create:guide -- --email ...	Provision guide user
Ingest DNR data	npm run ingest:dnr	Pull DNR lake data

Scripts live in scripts/.

6. Deploy & hosting

Action	Command / place
Local app	npm run dev
Deploy website	npm run deploy:hosting
Deploy functions	npm run deploy:functions
Rules / indexes / storage	firebase deploy --only firestore:rules,firestore:indexes,storage
Firebase project pointer	.firebaserc -> castiq-app
Hosting / emulator config	firebase.json

Typical live URLs (confirm in Firebase Console -> Hosting):

- https://castiq-app.web.app
- https://castiq-app.firebaseapp.com

7. Accounts, logins & secrets

Real passwords and API keys are NOT stored in this repository (by design). They live in vendor dashboards or Firebase Secrets. Share this PDF freely; share actual secrets only via a password manager - never in GitHub, email, or Slack.

App user logins (Firebase Auth)

Item	Details
How users log in	Email + password on /login
Where accounts live	Firebase Console -> castiq-app -> Authentication
Who can create accounts	Anyone via signup in the app
Guide accounts	Created with npm run create:guide (sets role: guide)
README example (template only)	guide@example.com / TempPass123! - only if that script was run
Script default if password omitted	CastIQ-Guide-ChangeMe1 (change immediately)
Password reset	Forgot password on login -> Firebase email reset

There is no shared master app username/password in the codebase. Each person has their own Firebase Auth user.

External software / API access

Service	Used for	Where credentials live	Login
GitHub	Source code	Repo access for swehrcustoms/CastIQ-App	github.com
Firebase / Google Cloud	Auth, DB, hosting, functions	Console project castiq-app	console.firebase.google.com

Service	Used for	Where credentials live	Login
Mapbox	Lake map	VITE_MAPBOX_ACCESS_TOKEN in .env / build secrets	account.mapbox.com
Anthropic (Claude)	AI fishing intel	Firebase Secret ANTHROPIC_API_KEY (server only)	console.anthropic.com
Tomorrow.io	Weather	Firebase Secret TOMORROW_IO_API_KEY	Tomorrow.io dashboard
Fishbrain / Navionics	Optional partners	Partner API keys in env (if enabled)	Partner portals
Firebase Admin (local seeds)	Running seed scripts	serviceAccountKey.json (never commit)	Firebase -> Service accounts

Frontend env template: .env.example -> copy to .env (gitignored).

Set Cloud Function secrets with:

```
firebase functions:secrets:set ANTHROPIC_API_KEY
firebase functions:secrets:set TOMORROW_IO_API_KEY
```

8. "I need to change X - where do I go?"

Goal	Go here
Change UI / screens	src/pages/ + src/components/
Change colors / look	tailwind.config.cjs
Change login flow	src/pages/Login.tsx, src/services/auth.ts
Change map	src/components/map/MapView.tsx + Mapbox token
Change AI behavior	functions/src/ai/ + Anthropic key
Change weather	functions/ weather/cache paths + Tomorrow.io key
Add/edit lake data	Firestore lakes or scripts/seedLakes.ts
Change who can read/write data	firestore.rules
Add a guide business	npm run create:guide + Firebase Auth
Deploy a new website build	npm run deploy:hosting
Deploy backend logic	npm run deploy:functions
Invite a business partner to code	GitHub repo settings -> Collaborators
Invite to Firebase / keys	Firebase Console -> Users & permissions

9. Credentials checklist for the partner

Hand these off outside this document (password manager):

- 1. GitHub invite (or org access) to CastIQ-App
- 2. Firebase / Google Cloud access to project castiq-app
- 3. Mapbox account (or token)
- 4. Anthropic API key (Firebase secret, not frontend)
- 5. Tomorrow.io API key (Firebase secret)
- 6. Any seeded test accounts (email/password) that were actually created
- 7. Optional: service account JSON for local seeding (keep offline)

This overview matches the CastIQ-App repository layout. Secrets are intentionally omitted. For the latest code and deploy steps, see README.md in the repo.